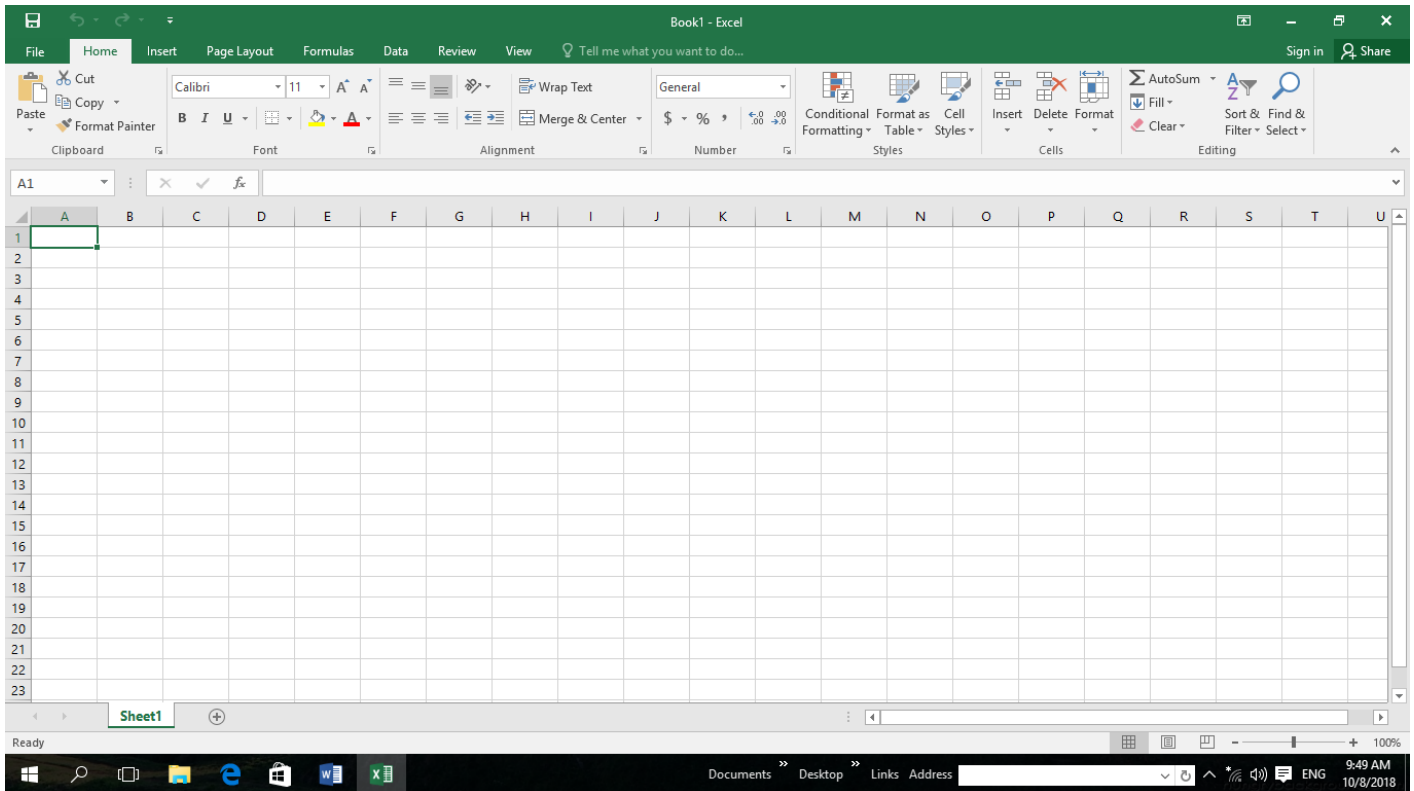


Spreadsheets - Microsoft Excel

Introduction:

Microsoft Excel is an electronic spreadsheet. You can use it to organize your data into rows and columns. You can also use it to perform mathematical calculations quickly.



Work Book:

A Microsoft Excel file is called a **Work Book**.

Worksheets:

A **Work Book** consists of several **Worksheets**. Each worksheet contains columns and rows. The letters are named A to Z and then continuing with AA, AB, AC and so on. The rows are numbered 1 to 1,048,576. The combination of a column coordinate and a row coordinate makeup a **Cell**. All these cells have **Cell Addresses**. For Example, the cell located in the upper left corner of the worksheet is cell **A1**, meaning **Column A** and **Row 1**. You enter your data into the cells on the worksheet.

Formula Bar:

The cell address of the cell you are in, displays in the **Name Box** which is located on the left side of the **Formula Bar**. Cell entries display on the **Formula Bar**.

Create a New Workbook:

1. Open Excel → **Blank Workbook**
2. **File** → **New** → **Blank Workbook**
3. Press **Ctrl + N**

Move Around a Worksheet:

Key Stroke	Action
Down arrow	Move downwards; one cell at a time
Up arrow	Move upwards; one cell at a time
Right arrow	Move right; one cell at a time
Left arrow	Move left; one cell at a time
Tab	Move right; one cell at a time
Shift + Tab	Move left; one cell at a time
Page Up / Page Down	Move up / down; one page at a time
Ctrl + Home	Move to the beginning of the worksheet
Ctrl + Right	Move to the right end of the same row
Ctrl + Left	Move to the left end of the same row
Ctrl + Down	Move to the bottom end of the same column
Ctrl + Up	Move to the top end of the same column
F5	Move from the current cell to another specific cell you required
Ctrl + G	
Type the cell address in the Name Box	
Home tab → Editing group → Go to	

Format Worksheet Tabs:

Renaming a Worksheet:

1. Open the sheet to be renamed → **Home tab → Cells group → Format → Rename Sheet** → Type a new name → Press **Enter** key.
2. Right click on the current sheet name → **Rename** → Type a new name → Press **Enter** key.
3. Double click on the current sheet name → Type a new name → Press **Enter** key.

Change Sheet Tab Color:

1. Open the sheet → **Home tab → Cells group → Format → Tab Color** → Select any color
2. Right click on tab → **Tab Color** → Select any color

Repositioning Worksheets:

Click and hold the worksheet tab that will be moved, until an arrow appears on the left corner of sheet tab → Drag the worksheet to the desired location → Release the mouse button

Insert and Delete Worksheets:

Insert Worksheets:

1. **Home** tab → **Cells** group → **Insert** → **Insert Sheet**
2. Right click on a sheet tab → **Insert** → **Work Sheet** → **Ok**
3. Click on the ⊕ symbol to the right of the sheet tabs

Delete Worksheets:

1. **Home** tab → **Cells** group → **Delete** → **Delete Sheet**
2. Right click on the sheet tab → **Delete**

Copy and Move Worksheets:

Copy Worksheets:

1. Right click on the sheet tab → **Move or Copy** → If you want to copy the sheet to another workbook, select the workbook you want from **To Book** → **Before Sheet** → Select the option you want → Select (check) **Create a Copy** → **Ok**
2. **Home** tab → **Cells** group → **Format** → **Move or Copy Sheet** → If you want to copy the sheet to another workbook, select the workbook you want from **To Book** → **Before Sheet** → Select the option you want → Select (check) **Create a Copy** → **Ok**

Move Worksheets:

1. Right click on the sheet tab → **Move or Copy** → If you want to move the sheet to another workbook, select the workbook you want from **To Book** → **Before Sheet** → Select the option you want → **Ok**
2. **Home** tab → **Cells** group → **Format** → **Move or Copy Sheet** → If you want to move the sheet to another workbook, select the workbook you want from **To Book** → **Before Sheet** → Select the option you want → **Ok**

Select Cells:

1. By clicking the mouse and dragging it.
2. By pressing the arrow keys while holding down the **Shift** key.
3. Holding down the **Ctrl** key and clicking the cells you want enables you to select noncontiguous areas of the worksheet.
4. To select a row or a column click on the row or column header.
5. To select the whole spreadsheet, click on the triangular shape before column A.
6. Click on a cell → Press **F8** key. This anchors the cursor. Note that “**Extend Selector**” appears on the status bar in the lower left corner of the window. You are in the **Extend Mode** → Click on another cell → Excel highlights the range. (Press **Esc** key and click anywhere on the worksheet to clear the highlighting)

To Edit a Cell:

Click on the cell →

1. Press **F2**
2. Use the formula bar
3. Double click on the cell

Saving a Workbook:

Saving a File for the First Time:

1. **File** tab → **Save / Save As** → Select the location you want to save → Type a **File Name** → **Save**
2. **Ctrl + S** → Select the location you want to save → Type a **File Name** → **Save**
3. Click on the **Save** button on the Quick Access tool bar → Select the location you want to save → Type a **File Name** → **Save**

After saving for the first time, if you edit the workbook once again, to save the changes you have done,

1. **File** tab → **Save**
2. Just click on the **Save** button on the Quick Access tool bar
3. **Ctrl + S**

Saving a Copy with a Different Name / Different Location / Both:

File tab → **Save As** → Select the location you want to save → Type a **File Name** → **Save**

Saving with a Password:

File tab → **Save As** → Select the location you want to save → **Save As** → **Tools** → **General Options** → Type a **Password to open / Password to modify** → **OK** → **Reenter Password to Proceed** → Type the password → **OK** → **Save**

Removing the Password:

File tab → **Save As** → Select the location you saved the workbook → **Save As** → **Tools** → **General Options** → Delete the given **Password to open / Password to modify** → **OK** → **Save** → **Do you want to replace it?** → **Yes**

Open a Workbook:

1. **File** tab → **Open** → Browse for the location and select the file
2. **Ctrl + O** → Browse for the location and select the file

Close a Workbook:

File tab → Close

Exit from Excel:

1. Click the **Close** button on the title bar
2. Right click on the title bar → **Close**

Editing a Worksheet:

Move and Copy Data:

1. Select the cell/s to be copied or moved → **Home** tab → **Clipboard** group → **Cut / Copy** button → Go to the new location → **Home** tab → **Clipboard** group → **Paste** button
2. Select the cell/s to be copied or moved → Right click on the selection → **Cut / Copy** → Right click on the new location → **Paste**
3. Select the cell/s to be copied or moved → Click on the cell pointer with the left mouse button → Drag the selection on to another location and release the mouse button → The cells will be moved
4. Select the cell/s to be copied or moved → Click on the cell pointer with the left mouse button → Drag the selection on to another location while pressing the **Ctrl** key and release the mouse button → The cells will be copied
5. Select the cell/s to be copied or moved → Click on the cell pointer with the right mouse button → Drag the selection on to another location and release the mouse button → **Copy Here / Move Here**
6. Select the cell/s to be copied or moved → Click on the fill handle → Drag the selection to the right of the selected cells and release the mouse button → The selection will be copied to the adjacent cells
7. Select the cell/s to be copied or moved → **Ctrl + C / Ctrl + X** → Click on the new location → **Ctrl + V**

Auto Fill:

Select the cell/s you want → Click on the **Fill Handle** → Drag the **Fill Handle** to complete the cells

Insert Cells:

1. Select the cell/s → **Home** tab → **Cells** group → **Insert** → **Insert Cells** → Select the option you want → **OK**
2. Select the cell/s → Right click on the selected area → **Insert** → Select the option you want → **OK**

Insert Rows / Columns:

1. Select the row/s or column/s → **Home** tab → **Cells** group → **Insert** → **Insert Sheet Rows / Insert Sheet Columns**

2. Select the row/s or column/s → Right click on the selected area → **Insert**

Delete Cells:

1. Select the cell/s you want to delete → **Home** tab → **Cells** group → **Delete** → **Delete Cells...** → Select the option you want → **OK**
2. Select the cell/s → Right click on the selected area → **Delete** → Select the option you want → **OK**

Delete Rows and Columns:

1. Select the row/s or column/s you want to delete → **Home** tab → **Cells** group → **Delete** → **Delete Sheet Rows / Delete Sheet Columns**
2. Select the row/s or column/s you want to delete → Right click on the selected area → **Delete**

Hide Rows / Columns / Sheets:

1. Select the row/s or column/s or Sheet/s you want to hide → **Home** tab → **Cells** group → **Format** → **Hide & Unhide** → Choose the option you want
2. Select the row/s or column/s or Sheet/s you want to hide → Right click on the selected area → **Hide**
3. To hide a sheet, right click on the sheet name → **Hide**

Unhide Rows / Columns / Sheets:

1. Select the row/s or column/s which are adjacent to the hidden row/s or column/s → **Home** tab → **Cells** group → **Format** → **Hide & Unhide** → **Unhide Rows / Unhide Columns**
2. Select the row/s or column/s which are adjacent to the hidden row/s or column/s → Right click on the selected area → **Unhide**
3. To unhide a sheet, right click on the sheet name → **Unhide** → Select the sheet name → **OK**

Format Cells:

1. Select the cell/s you want to format → Click on the launcher of **Font / Alignment / Number** group → You will open the **Format Cells** dialog box → Click on any tab → Use the given tools to format data
2. Select the cell/s you want to format → Right click on the selected area → **Format Cells** → You will open the **Format Cells** dialog box → Click on any tab → Use the given tools to format data
3. Select the cell/s you want to format → **Home** tab → **Font / Alignment / Number** group → Use the given tools to format data

Change Column Width & Row Height:

1. Keep the mouse pointer on the column header or row header → Use  symbol to drag the line in or out
2. Select the cell / row / column → **Home** tab → **Cells** group → **Format** → **Row Height / Column Width** → Enter the value you want → **OK**

Excel Formulas:

A **Formula** is a set of mathematical instructions that can be used in Excel to perform calculations. Formulas are started with a = sign.

E.g.:

=A1 + B2	=G3-H6	=A1*B1	=A1/B3
=D2+D3+D4+D5	=B5-A2	=C2*D2*E4*F5	=A3/C1
=45+56+87	=67-34	=23*5	=75/5

Excel Functions:

A **Function** is a built in formula in Excel.

Function	Description	Example
Sum	Adds all cells in the argument	=Sum(A2:A15)
Average	Calculate the average of the cells in the argument	=Average(C5:C15)
Min	Finds the minimum value in the range	=Min(A4:A20)
Max	Finds the maximum value in the range	=Max(D6:F12)
Count	Finds the number of cells that contain a numerical value in the range	=Count(A1:A10)
Counta	Finds the number of cells that contain any type of data in the range	=Counta(C1:E10)
If	Gets the logical answer according to a given statement	=If(D3>50,"Pass","Fail")
Rank	Finds the position of a value either in ascending order or descending order according to a given range of values	=Rank(F8,\$F\$8:\$F\$12,0)

Apply Functions:

1. Keep the cell pointer on the cell where you want to write the function → Type the function → Press **Enter** key
2. Keep the cell pointer on the cell where you want to write the function → **Formulas** tab → **Function Library** group → **Insert Function** → Select the function → **OK** → Complete the **Function Arguments** dialog box → **OK**

3. Keep the cell pointer on the cell where you want to write the function → **Formulas** tab → **Function Library** group → **Auto Sum** → Select the function you want → If the highlighted range is not relevant, highlight the relevant range → Press **Enter** key
4. Keep the cell pointer on the cell where you want to write the function → **Home** tab → **Editing** group → **Auto Sum** → Select the function you want → If the highlighted range is not relevant, highlight the relevant range → Press **Enter** key
5. Keep the cell pointer on the cell where you want to write the function → Click on the **Insert Function (fx)** button → Select the function you want → **OK** → Complete the **Function Arguments** dialog box → **OK**
6. Type the = sign in the cell where you want to write the function → Click on the drop down button of the Name box → Select the function you want → Complete the **Function Arguments** dialog box → **OK**

Cell References:

There are two types of cell references: **Relative** and **Absolute**. Relative and absolute references behave differently when copied and filled to other cells. Relative references **change** when a formula is copied to another cell. Absolute references, on the other hand, remain **constant**, no matter where they are copied.

Relative references:

By default, all cell references are **relative references**. When copied across multiple cells, they change based on the relative position of rows and columns. For example, if you copy the formula **=A1+B1** from row 1 to row 2, the formula will become **=A2+B2**.

Absolute references

There may be times when you do not want a cell reference to change when filling cells. Unlike relative references, **absolute references** do not change when copied or filled. You can use an absolute reference to keep a row and/or column **constant**. An absolute reference is designated in a formula by the addition of a **dollar sign (\$)**. It can precede the column reference, the row reference, or both.

\$A\$2	The column and the row do not change when copied
A\$2	The row does not change when copied
\$A2	The column does not change when copied

Sorting Data:

1. Select the data range you want to sort → **Home** tab → **Editing** group → **Sort & Filter** → **Sort Smallest to Largest / Sort Largest to Smallest / Sort A to Z / Sort Z to A** → If you receive a **Sort Warning**, select the option you want → **Sort**
2. Select the data range you want to sort → Right click on the selected range → **Sort** → **Sort Smallest to Largest / Sort Largest to Smallest / Sort A to Z / Sort Z to A** → If you receive a **Sort Warning**, select the option you want → **Sort**
3. Select the data range you want to sort → **Home** tab → **Editing** group → **Sort & Filter** → **Custom Sorts** → If you receive a **Sort Warning**, select the option you want → **Sort** → Choose which column you want to sort by first → Click **Add Level** → Choose the next column you want to sort → **OK**

Filtering Data:

1. Select the data range you want to filter → **Home** tab → **Editing** group → **Sort & Filter** → **Filter** → Click on the drop down button of the first cell in the highlighted data range → Filter data by using suitable options
2. Select the data range you want to filter → Right click on the selected range → **Filter** → Select any option

Apply the Advanced Filter:

Set up the Criteria Range -

In the criteria range for an Excel advanced filter, you can set the rules for the data that should remain visible after the filter is applied. You can use one criterion, or several.

- ☐ In this example, cells H1:H2 are the criteria range.
- ☐ The heading in H1 should exactly matches a heading in the database.
- ☐ Cell F2 contains the criterion. The > (greater than) operator is used, with the number 500 (no currency sign is included).

H
Total
<500

Type the criteria in a separate cell. → Select a cell in the database → **Data** tab → **Sort & Filter** group → **Advanced** → You can choose to filter the list in place, or copy the results to another location → Excel should automatically detect the list range. If not, you can select the cells on the worksheet → Select the criteria range on the worksheet → If you are copying to a new location, select a starting cell for the copy → **OK**

Charts:

Create a Chart –

Select the cells that contain the data you want to use in the chart → **Insert** tab → **Charts** group → Select the chart type you want

Add Chart Elements –

Select the chart → **Chart Tools** tab → **Design** tab → **Chart Layouts** group → **Add Chart Elements** → Select any option and add elements

Move a Chart –

Select the chart → **Chart Tools** tab → **Design** tab → **Location** group → **Move Chart** → Choose where you want the chart to be placed → **OK**

Use the tools in **Design** and **Format** tabs to do the changes you want to your chart.

Set Page Layout of a Worksheet:

1. **Page Layout** tab → Click on the **Page Setup** tab launcher → Use **Page**, **Margins**, **Header / Footer** and **Sheets** tabs to arrange the page setup → **OK**
2. **Page Layout** tab → Use the tools in **Page Setup** group to arrange the page setup

Set Print Area:

Select the area you want to print → **Page Layout** tab → **Page Setup** group → **Print Area** → **Set Print Area**

Printing a Worksheet:

1. After arranging the page setup in the **Page Setup** dialog box, Click **Print** → If you want, make necessary changes there → **Print**
2. **File** → **Print** → If you want, make necessary changes there → **Print**